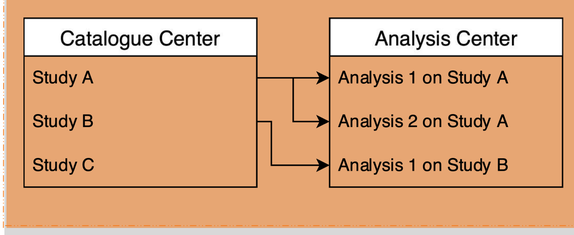
Federated Learning For Everyone (FL4E)

ELIXIR All Hands , 5-8 June 2023, Dublin, Ireland

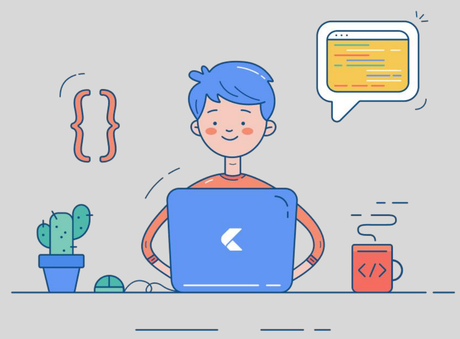
As healthcare data grows more abundant and clinical research questions become more complex, it is crucial for clinical data collection initiatives and data analysis experts to collaborate effectively. Federated learning ecosystems allow for multiple data sources to contribute to statistical model learning without compromising patient privacy and anonymity.
Current lack of standard architecture leads to recurrent costs for designing dedicated federated solutions for each initiative. Our customizable data analysis platform addresses various data sharing requirements encountered in clinical practice.

Study Center

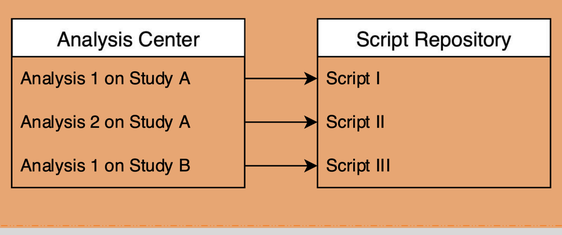
•The Study Center allows users to initiate and manage a catalog of their studies.
•Users can create specific analyses and assign them to individuals or groups, controlling authorization and access levels.



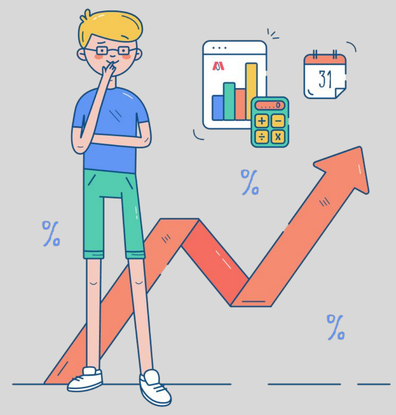
Repository Center



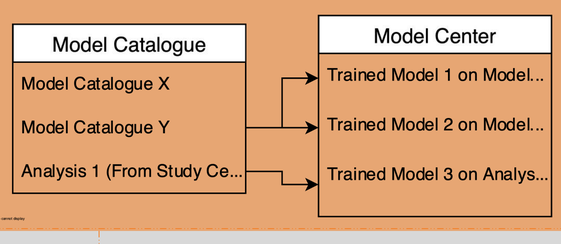
•The Repository Center serves as a warehouse, storing all scripts and models associated with study catalogs.
•This module centralizes resources and ensures efficient access and management of study materials.



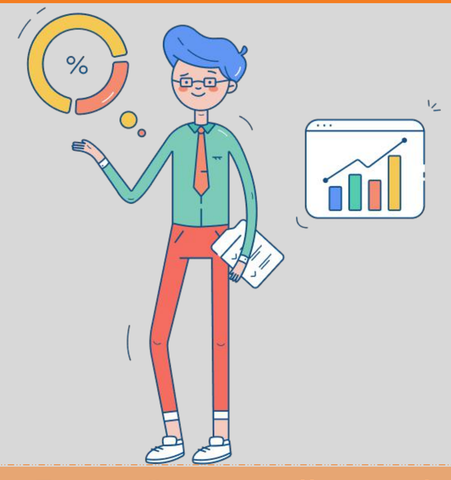
Model Center



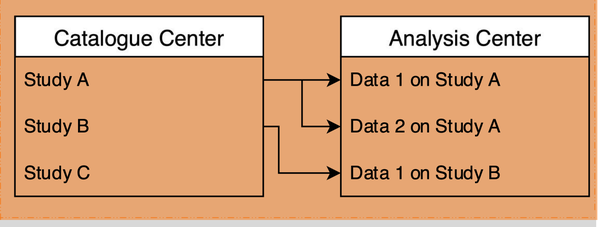
•The Model Center holds trained models resulting from studies or other external sources.
•This module enables integration of a decision support system with trained models to enhance data-driven decision-making.



Data Center



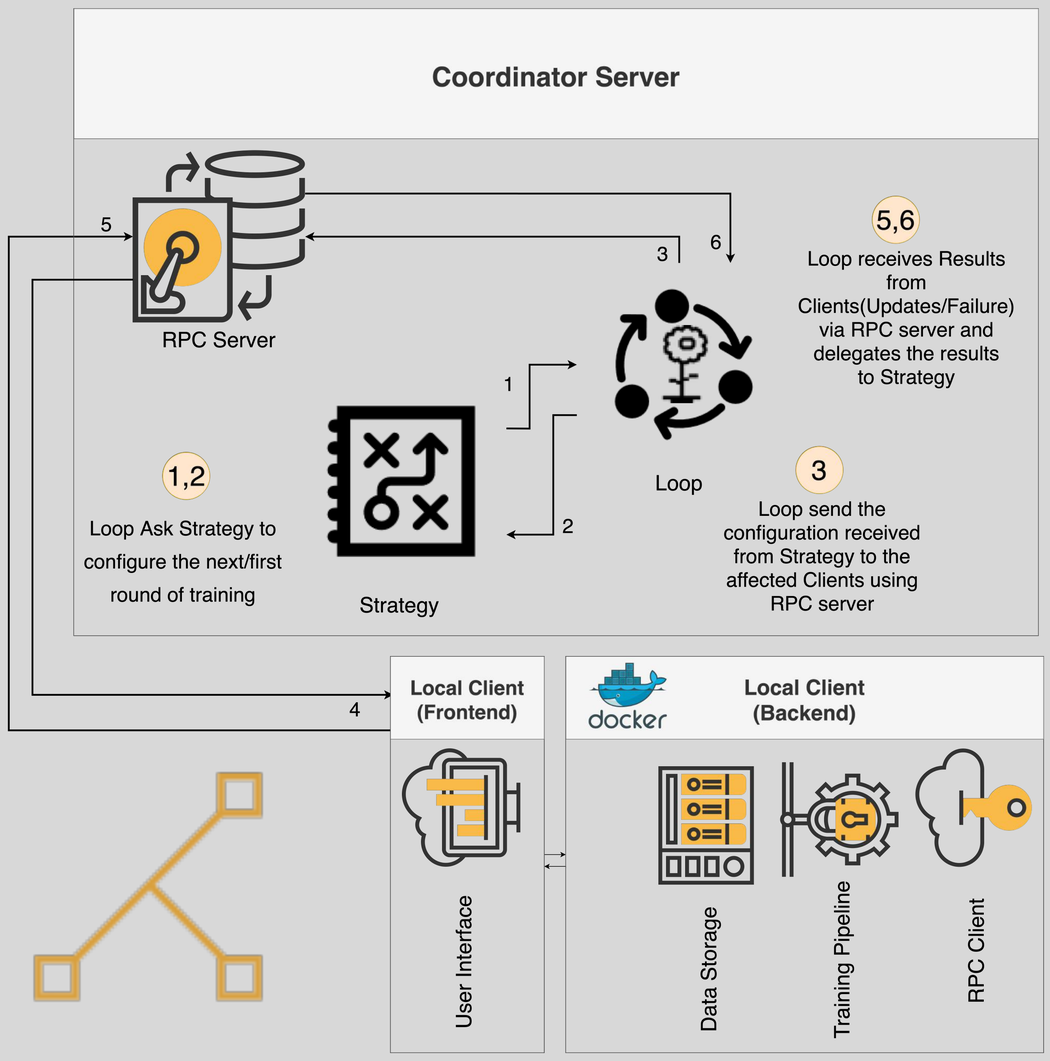
•The Data Center allows data custodians or individuals to share patient-level data, subject to terms and conditions, with studies.
•This module ensures inclusiveness for users who may not want to join as federated clients but still want to contribute data.



Versatile Federated Ecosystem for Multi-stakeholders Clinical Research Collaboration:

FL4E infrastructure proposed here follows a nested approach, as illustrated in each module's figure. The infrastructure can accommodate multiple ecosystems, which can be understood as general data sharing and clinical research initiatives.
FL4E utilizes the Friendly Federated Learning Framework[1] as its backend, offering an inclusive and robust foundation for the customizable data analysis platform. This ML framework-agnostic backend supports federated analysis execution, providing a comprehensive solution for multi-stakeholder clinical research collaboration based on the 4 modules within the platform.

A high-level overview of federated analysis as a result of initiating a study center is presented. Addressing key challenges in data sharing, transparency, and ease of use, the framework encourages inclusiveness and effective collaboration among diverse stakeholders, with the potential to transform the clinical research landscape.
A prototype of this framework has been deployed for demonstration purposes, with all materials available in the GitHub repository. Access the demo using the following link: https://fl-server.azurewebsites.net.



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